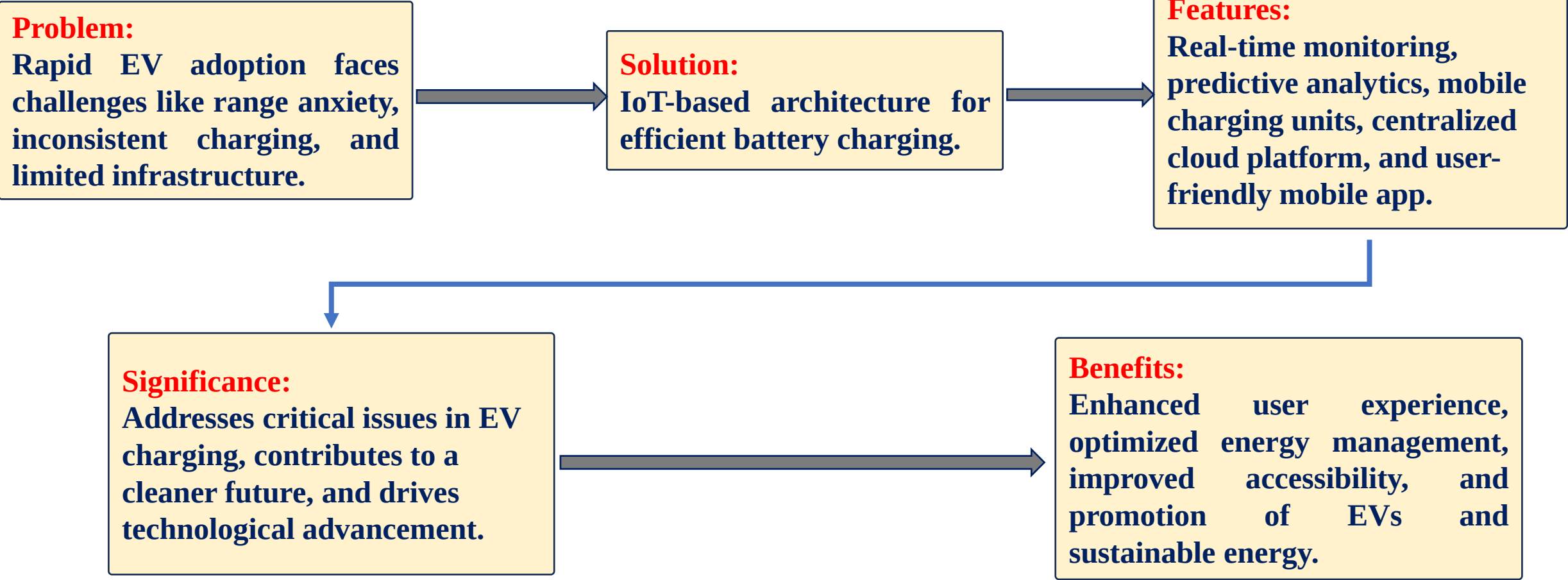


Project Title: **Design of IOT Architecture for Battery Charging Management System**

Theme of the Project: **Battery Swapping station and EV Charging Infrastructure**



Abstract



Introduction

- Rapid growth of electric vehicles (EVs) increasing demand for efficient charging infrastructure.
- Need for sustainable solutions integrating renewable energy sources like solar power and energy storage systems (ESS).

Objective:

Develop an IoT-based architecture to optimize battery charging management. Utilize real-time data monitoring and predictive analytics to enhance efficiency.

Key Benefits:

Significant reduction in operational costs, achieving up to 88.5% cost efficiency. Enhanced battery management through real-time monitoring of critical parameters (voltage, temperature, SOC).



Market Demand:

Increasing EV sales necessitating more charging stations. Consumer preference shifting towards renewable energy-powered solutions.

Future Scope:

Optimize operations and automate maintenance. Expand charging infrastructure to meet growing demand.

Motivation

Motivation, Research Gap & Problem, Novelty

- The rapid growth of electric vehicles (EVs) necessitates a robust and efficient charging infrastructure.
- Integrating renewable energy sources like solar power into charging systems aligns with environmental sustainability goals.
- The need to reduce operational costs in charging management drives innovation with IOT in battery management technologies.

Research Gap & Problem

- Absence of uniform standards for EV charging stations leads to compatibility issues across different systems.
- More research is needed to understand consumer preferences and behaviors regarding EV charging solutions.
- Limited Integration of Renewable Energy Sources
- Absence of Comprehensive IoT-Based Charging Management
- No Holistic Approach to Grid Stability
- Minimal Real-Time Monitoring and Control Capabilities



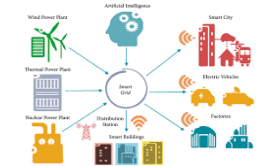
IOT
Connectivity



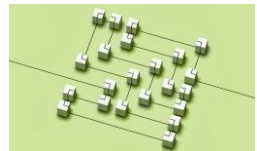
Real Time
monitoring



Load management



Grid Stability



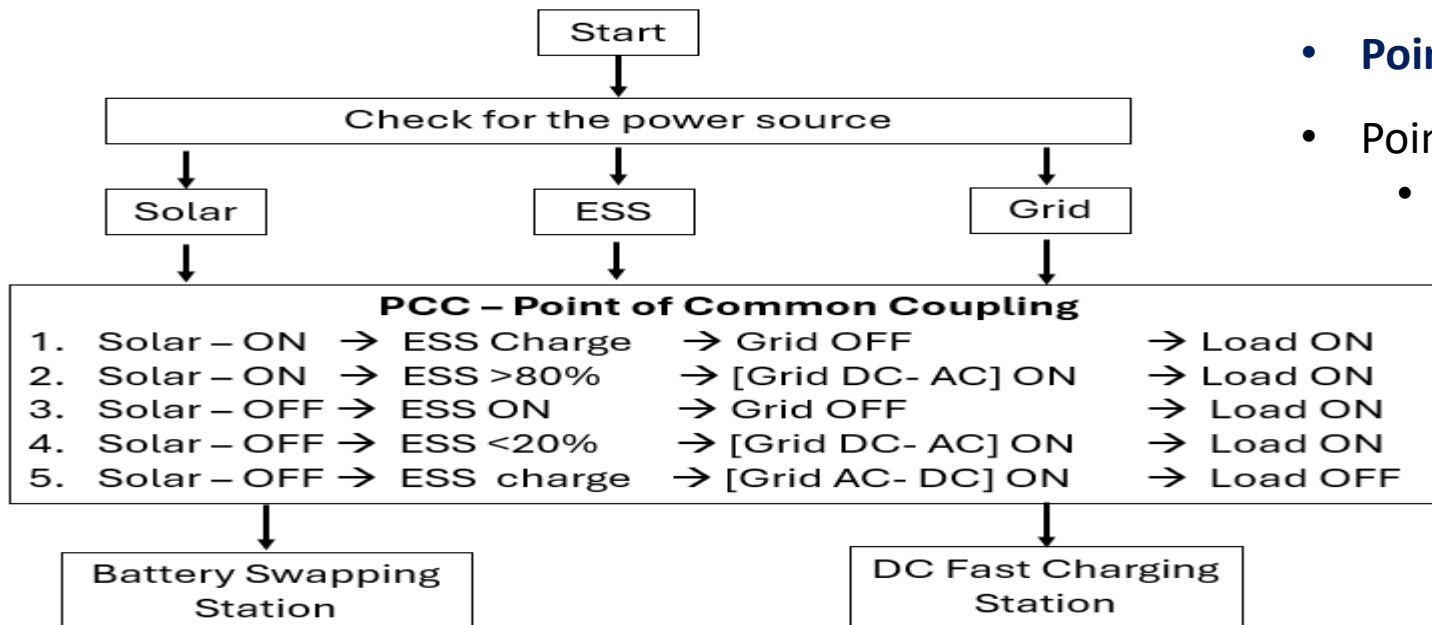
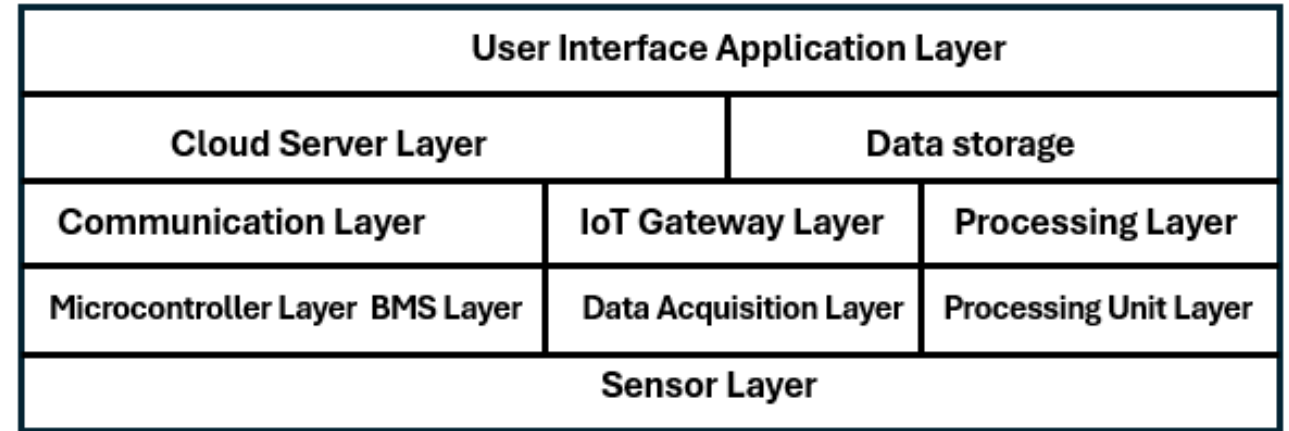
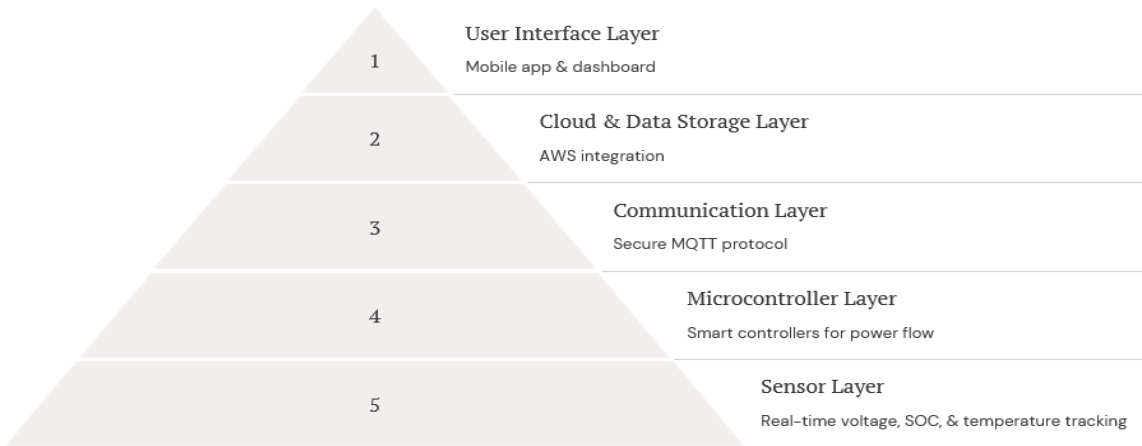
Algorithm

Novelty

- The proposed architecture uniquely combines IoT technology for real-time monitoring in battery charging management.
- The five-layered architecture and PCC with source integration and automation provides a comprehensive approach to optimize charging processes.
- Preliminary simulations indicate a potential reduction of up to 88% in operational costs through optimized resource utilization.

Methodology

This methodology aims to create a comprehensive IoT architecture that enhances battery charging management while promoting sustainability through the integration and Automation of renewable energy sources with Realtime monitoring



- **Point of Common Coupling - PCC**
- Point of Common Coupling for integration and Automation
 - Combines solar power, Energy Storage Systems (ESS), and grid power at the Point of Common Coupling.

Methodology

Algorithms for Proposed Methodology

MPPT (Maximum Power Point Tracking) Algorithm:

- Optimizes solar panel output by continuously adjusting the operating point to maximize energy capture under varying conditions.

Battery Swapping Station Algorithm:

- Manages logistics and scheduling for battery swaps, optimizing inventory and availability of charged batteries to improve service efficiency.

Vehicle Charging Control Calculation:

- Calculates optimal charging rates based on battery state-of-charge (SOC) and user demand, ensuring safe and efficient charging.

Safety and Blast Condition Prediction using Machine Learning:

- Utilizes machine learning models to analyze historical data and predict potential safety hazards in battery charging environments, enabling proactive risk management.

AWS Cloud Integration Methodology

Secured MQTT Protocol:

- Utilizes MQTT for secure and efficient communication between IoT devices and the cloud.

S3 Bucket Data Storage:

- Stores collected data in Amazon S3 for scalable and durable storage solutions.

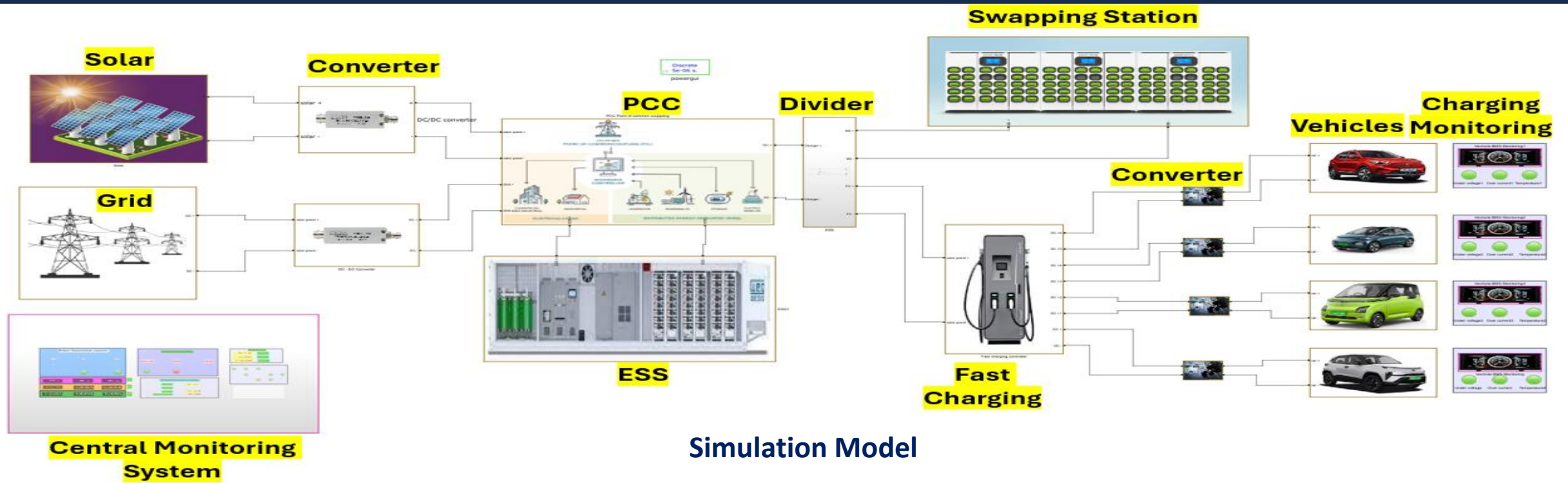
Email Alerts:

- Implements automated email notifications for critical events or alerts related to battery charging management.

Data Visualization:

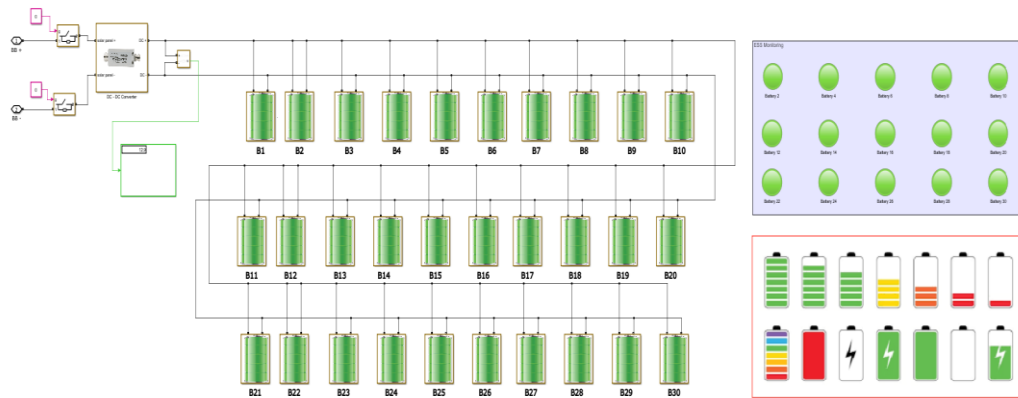
- Provides real-time data visualization dashboards to monitor charging status and performance metrics.

Results –Simulation models

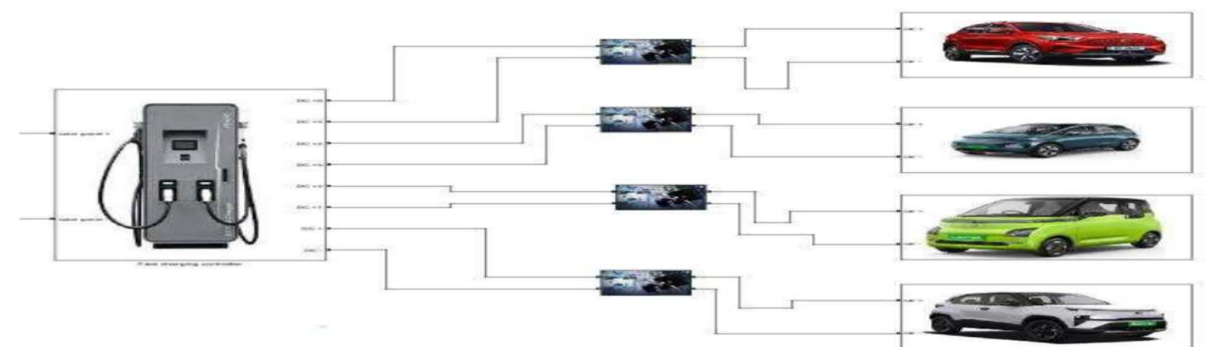


Simulation Model

Battery Swapping station



EV Charging

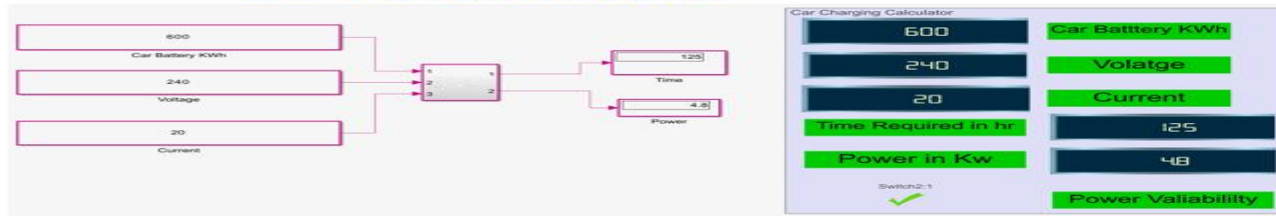


Results

Central Monitoring Station



EV Parameters



EV Battery charging Parameters



Visualization - Monitoring



AWS Cloud Connectivity

Estimated Energy and Cost calculation

Power Generation

Power Source	Voltage [V]	Current [I]	% of Generation/day	Power [P]- KWh	Power/day
Solar	430	100	60	43	1032
ESS	480	50	30	24	576
Grid	435	20	10	9	209
Total Power Generation per day in KW/24 hr					1817

PCC - Point of common coupling

PCC will Logically be controlled on availability of source		
Voltage - [V]	430	Output Power/day in KW
Current - [I]	80	
Power - [W] - KW/h	75	
Power Availability	1.17	1772 use / day

ESS - Charging

No.of batteries	40
Voltage	12
Amper hour - capacity	100
Power / charge - Watt / hr	48000
Power / one cycle charge - KW	48

Fast Charging Station Load

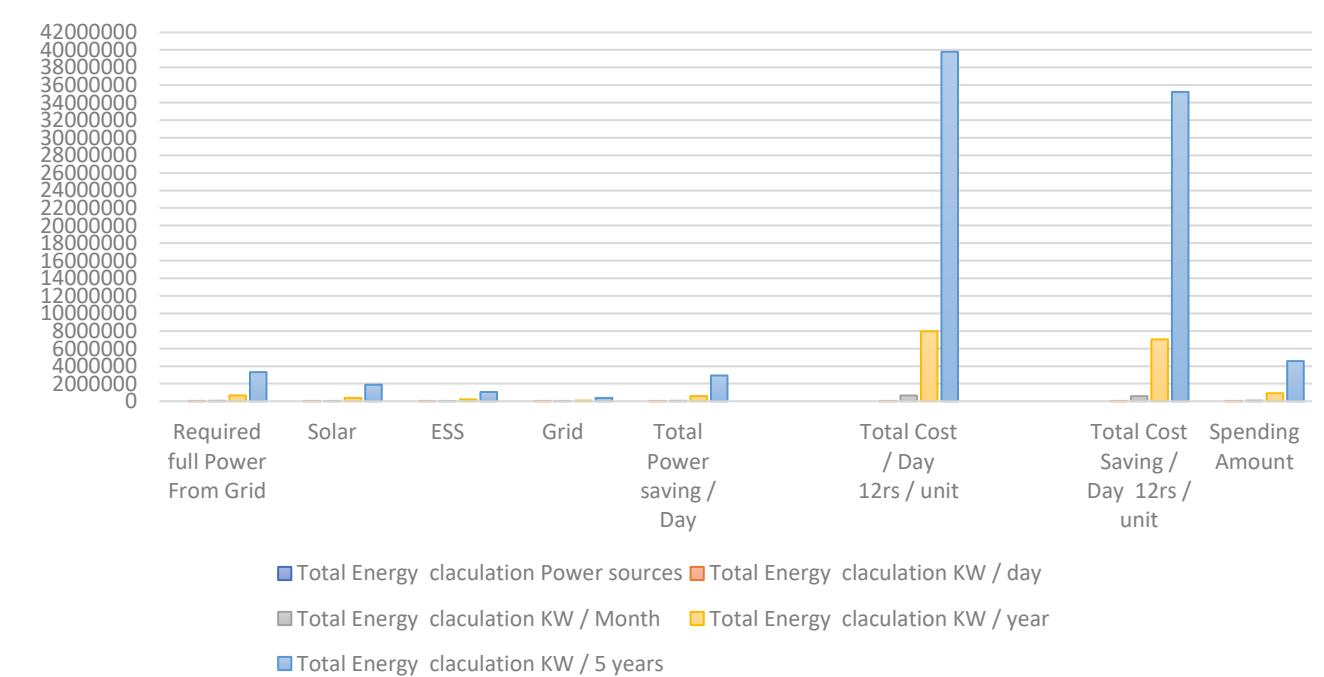
Cars	Voltage (V)	Power (kW)	Battery Capacity (kWh)	SOC Available	Required Current	Charging Time in hr	Unit charge	Cost
1 - Tata Nexon - EV	400	30.2	30.2	20	75.5	2.47	10	745.83
2- MG ZS EV	400	50	44.5	30	125	1.49	10	743.75
3-Tata Trigo EV	400	16.8	26	50	42	4.40	10	739.58
4- Mahindra eVerito	400	70	60	10	175	1.07	10	747.92
Total Power		167			Total Time	9	Total Revenue	2977

Battery Swapping Station

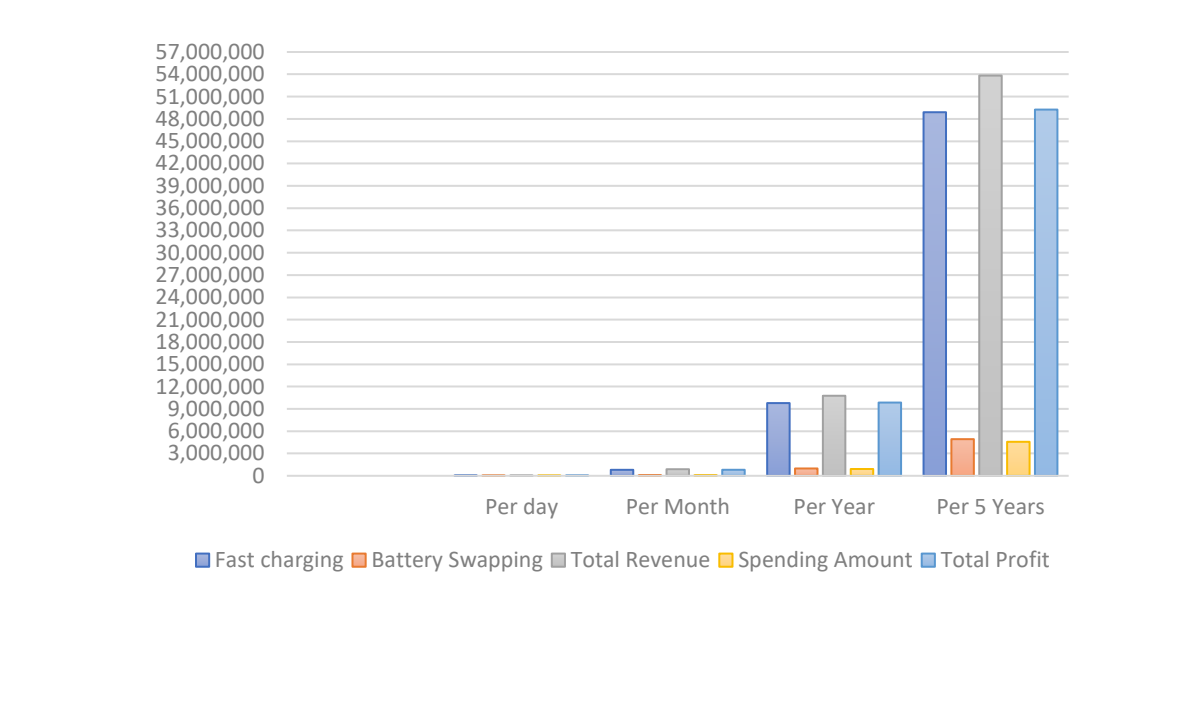
Battery Swapping Station									
Total Batteries	30	<table><tr><td>Total Batteries charging / day</td><td>90</td></tr><tr><td>Cost of charge per battery</td><td>30</td></tr><tr><td>Total Amout Per day</td><td>2700</td></tr></table>	Total Batteries charging / day	90	Cost of charge per battery	30	Total Amout Per day	2700	
Total Batteries charging / day	90								
Cost of charge per battery	30								
Total Amout Per day	2700								
Under maintance	0								
Avaliable batteries	30								
Voltage	12								
Current	10								
Time / Battery in hr	5								
No of Cycle per day	3								
Power/Battery - KW	0.12								
Total Station KW	18								
Total Energy Required for Full Swapping Station per Day - KW	54								

Total Power	1670	KW power can charge / day	Total Time	94 / 96 hr	Total Revenue	26794
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Total Energy calculation				
Power sources	KW / day	KW / Month	KW / year	KW / 5 years
Required full Power From Grid	1,817	54,504	663,132	3,315,660
Solar	1,032	30,960	376,680	1,883,400
ESS	576	17,280	210,240	1,051,200
Grid	209	6,264	76,212	381,060
Total Power saving / Day	1,608	48,240	586,920	2,934,600
Total Cost / Day 12rs / unit	21,802	654,048	7,957,584	39,787,920
Total Cost Saving / Day 12rs / unit	19,296	578,880	7,043,040	35,215,200
Spending Amount	2,506	75,168	914,544	4,572,720



Revenue				
Source	Per day	Per Month	Per Year	Per 5 Years
Fast charging	26,794	803,813	9,779,719	48,898,594
Battery Swapping	2,700	81,000	985,500	4,927,500
Total Revenue	29,494	884,813	10,765,219	53,826,094
Spending Amount	2,506	75,168	914,544	4,572,720
Total Profit	26,988	809,645	9,850,675	49,253,374



Conclusion

Conclusion

- The proposed IoT architecture effectively addresses the growing demand for efficient EV charging infrastructure.
- It promotes sustainability by integrating renewable energy sources, enhancing operational efficiency, and reducing costs.
- Real-time monitoring and predictive analytics contribute to improved battery management and a greener transportation ecosystem.

Patent

 Office of the Controller General of Patents, Designs & Trade Marks Department for Promotion of Industry and Internal Trade Ministry of Commerce & Industry, Government of India  INTELLECTUAL PROPERTY INDIA PATENTS DESIGNS TRADE MARKS GEOGRAPHICAL INDICATIONS	
Application Details	
APPLICATION NUMBER	202541009491
APPLICATION TYPE	ORDINARY APPLICATION
DATE OF FILING	05/02/2025
APPLICANT NAME	SRM UNIVERSITY
TITLE OF INVENTION	SYSTEM AND METHOD FOR MANAGING CHARGING INFRASTRUCTURE OF ELECTRIC VEHICLES THROUGH IoT
FIELD OF INVENTION	COMPUTER SCIENCE
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E-MAIL (UPDATED Online)	
PRIORITY DATE	
REQUEST FOR EXAMINATION DATE	05/02/2025
PUBLICATION DATE (U/S 11A)	14/02/2025

Thankyou